

# *DSAA2011 Machine Learning*

## Sample Questions

1. (5 points) Determine whether the statement is true or false, and provide a brief explanation.
  - (a) (5 points) An affine classifier is also a linear classifier.

2. (10 points) Let  $\mathbf{x} = (x_1, x_2)^\top \in \mathbb{R}^2$  and objective function

$$C(\mathbf{x}) = \mathbf{x}^\top \mathbf{x}. \tag{1}$$

- (a) (2 points) Find  $\nabla_{\mathbf{x}} C(\mathbf{x})$ , i.e., the gradient vector of  $C(\mathbf{x})$  regarding to  $\mathbf{x}$ .
  - (b) (7 points) To minimize  $C(\mathbf{x})$ , we consider the gradient descent (GD) algorithm with learning rate  $\eta = 100$ . We set the initiation point  $\mathbf{x}_0$  as  $(1, 1)$  and then using the GD algorithm to generate a sequence:  $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots)$   
Based on the description provided, write a full implementation of the gradient descent algorithm in Python or pseudocode.
  - (c) (1 point) What is the convergence criterion used in part (b)?

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| Note: the number and difficulty of the questions in the exam will differ from those in the sample. |
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